Is the Finite-Shadow Language a Faithful Compression of Collatz Dynamics?

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Abstract

A referee asks: *how do we know the finite-shadow language captures all arithmetic mechanisms available to Collatz, rather than only the mechanisms visible to the surviving candidates inside the framework?* We answer in four parts. (1) The language is *lossless*, not a projection: the valuation-word coding is a homeomorphic conjugacy of the 2-adic Syracuse system onto the full shift, so no mechanism is discarded at the symbolic level; what the language cannot express is exactly one thing — membership in $\mathbb{N} \subset \mathbb{Z}_2$ — and the program imports that through an explicit, enumerable list of archimedean mechanisms (clocks, counting, the drift identity, Baker). (2) Within the finite-modulus category the language is provably *complete*: every observable of a bounded orbit window modulo any Q coprime to 6 factors exactly through (one residue, the letters), and conditional on any word cylinder the residues at all odd places are exactly equidistributed — only the 2-adic coordinate feeds the word; every other finite place is a passenger. (3) The empirical hidden-channel test is null: multiplicative invariants (ω, squarefreeness, divisibility, primality) of deep $\{1, 2\}$-survivors match size-matched controls exactly. (4) The "reversed-flow" concern conflates presentation with content: the load-bearing theorems of the program are $\forall$-integer statements in Tao's direction; and the audit of arXiv:2603.11066 shows three independently built frameworks translating term-for-term into one another and hitting the *same* residual wall — the language-invariance one expects of a faithful compression, not the framework-relativity one would fear.

1 The question, made precise

Call an *arithmetic mechanism* any constraint usable in a proof: an observable of orbits together with a theorem about it. The framework under audit speaks the language of valuation words $b_n = v_2(3x_n + 1)$, finite character shadows, 2-adic alignments (clocks), and band counting. The referee's worry: perhaps Collatz dynamics supports mechanisms invisible in this language, so that narrowing the surviving profile *inside* the framework overstates progress. Faithfulness splits into three claims, each addressable: (a) the coding loses nothing; (b) within its declared categories the language is complete; (c) what lies outside is isolated explicitly, not silently compressed away. All verification counts refer to `search/faithfulness.py`.

2 Losslessness: conjugacy, not projection

Theorem 1 (Lossless coding)**.** *The word map $x \mapsto (b_n(x))_{n\geq 0}$ is a homeomorphic conjugacy from the accelerated Syracuse system on the odd 2-adics onto the full shift on $\{1, 2, 3, \dots\}^{\mathbb{N}}$, carrying Haar measure to the Bernoulli measure $\mathbb{P}[b = k] = 2^{-k}$. A length-$k$ prefix w pins exactly the residue class $x \bmod 2^{B_w+1}$ and nothing more. (Verified: all 4845 realizable length-4 words each occupy exactly one class, total mass $0.9987 \to 1$; letter marginals are exactly 2^{-k} by residue counting.)*

This is the Bernstein–Lagarias conjugacy in our coordinates; our cylinder lemma is its quantitative form. Consequence: *no* mechanism is lost in passing to the symbolic language — the 2-adic system and the word system are the same object. The language has exactly one blind spot, and it is not a mechanism but a *subset*: $\mathbb{N} \subset \mathbb{Z}_2$ (dense, measure zero, non-invariant). The framework does not pretend to express it symbolically; it imports it through four explicit interfaces, each an exact theorem: the ghost clocks (positivity caps $a \le \log_2(|2^K - 3^\ell|x + C)$), without-replacement counting (an acyclic orbit visits each integer once), the drift identity (exact, with error $\sum \log_2(1 + \frac{1}{3x_n})$ explicitly bounded), and the Baker/S-unit inputs. Any mechanism the language misses must live in this declared residual — which is precisely the part the program treats as open.

3 Completeness within the finite-modulus category

Theorem 2 (k-point shadow completeness)**.** *For any Q coprime to* 6 *and any k: along any orbit, $x_{n+i} \bmod Q = (3^i x_n + A_i(w))\, 2^{-B_i(w)} \bmod Q$ for $i \le k$, so* every *Q-periodic observable of a k-step window factors exactly through $(x_n \bmod Q,\ b_n, \dots, b_{n+k-1})$; and for every multiplicative character, $\chi(x_{n+1})\chi(2)^{b_n} = \chi(3x_n + 1)$ exactly — the keystone identity. (Verified:* 2000 *random windows, $Q \in \{5, 7, 11, 13, 25, 35\}$.)*

Theorem 3 (CRT orthogonality)**.** *Conditional on any word cylinder (an arithmetic progression modulo 2^{B+1}), the residues modulo any odd m are* exactly *equidistributed. Hence the word reads only the* 2*-adic coordinate of x; every other finite place evolves as an explicit skew product driven by the word and carries no information about future letters beyond it. (Verified:* 40 *random depth-*18 *cylinders, exact uniformity mod* $3, 5, 7$*.)*

Together: *within the category of finite-modulus observables of bounded windows — the natural home of "congruence mechanisms" — the shadow language is complete, provably and exactly.* A congruence mechanism unavailable in the language does not exist.

4 The hidden-channel test

Mechanisms outside the congruence category would have to use the multiplicative or global structure of orbit values (factorization, primality, smoothness). Theorem 3 does not force these to be neutral, so we test: among the 2800 odd $n_0 < 2^{22}$ surviving 23 letters in $\{1, 2\}$, against size-matched random odd controls:

	mean $\omega(n)$	$\mathbb{P}$[squarefree]	$\mathbb{P}[3 \mid n]$	$\mathbb{P}$[prime]
survivors	2.480	0.806	0.334	0.144
controls	2.478	0.815	0.319	0.144

(the squarefree density of *odd* integers is $8/\pi^2 = 0.811$; both populations sit on it). Null within sampling error: no multiplicative channel into survival is detected. A deviation here would have been a genuine mechanism outside the language; finding none is the falsifiable content of faithfulness, and the test is cheap to repeat at larger depth.

We record one instructive artifact: ensemble letter autocorrelation measured naively on 2^{42}-sized seeds gave $+0.021$, an apparent violation of the Bernoulli prediction — caused entirely by absorption into the trivial cycle (the constant $b = 2$ tail). Measured on pre-absorption windows it is -0.003 ± 0.003. The framework predicted both numbers; apparent unfaithfulness was, on inspection, the $+1$ ghost.

5 The direction of flow

The concern that the program only narrows a surviving profile — reversing Tao's all-integers flow — describes its *presentation*, not its theorems. The load-bearing results are universal statements over all integers or all acyclic orbits, in exactly Tao's direction:

∀-statement	form
narrow-band dichotomy	all orbits, dwell $\leq CY^{\theta(D)}$
logarithmic dwell at ratio ≤ 2	all acyclic orbits
escape rate $O(X \log X)$	all divergent orbits
upcrossing bound 2^{k-3}	all divergent orbits
two-clock identities, parse	all odd integers, exact
universal ghost clock, supply	all signatures, all orbits
exact counts $N_D(p)$, $(3/4)^D$ law	all words/classes
entropy deficit $\theta_{12} < 0.9791$	all bands, all widths
Haar i.i.d. round laws	all residue classes

The "surviving profile" is the *complement* of these universal constraints, stated for orientation. Compressing the complement is how universal theorems report progress.

6 Language-invariance of the residual

Finally, the strongest available evidence that the compression is canonical rather than self-serving: independent frameworks built in different languages translate into ours term by term. In the audit of arXiv:2603.11066: their burst–gap calculus *is* the two-clock parse (their $k = v_2(n+1)$ is our depth g; their Persistent Exit Lemma is our parity handoff at $h = 3$, verified 4000/4000); their mean-gap threshold 1.71 is our $1/\log_2(3/2)$; their two open hypotheses are our two calibration channels; their CIC is the parity-$\equiv 1$ extreme of our calibration conjecture; their phantom-cycle census is our ghost catalogue with the same roots $C_\sigma/(2^K - 3^\ell)$. Tao's framework (Syracuse random variables, 3-adic drivers, logarithmic density) likewise meets the same final gap — almost-all versus all, distributional versus pointwise, annealed versus quenched. Three languages, one wall: the obstruction is a property of the dynamics, not of any framework. Were the finite-shadow language an unfaithful narrowing, its wall would be an artifact located elsewhere than the others'; instead all three walls coincide under explicit dictionaries.

We also record, for honesty, where the framework's *stochastic heuristics* (as opposed to its language) are known to deviate from exact dynamics: the defect-cocycle C-suite measured exact-vs-annealed gaps (operator gaps, excursion-length-dependent shadow rates). None of the unconditional theorems above rely on annealed models; the measured gaps are themselves expressed — and were found — *inside* the language.

7 Does the blind spot itself constrain the profile?

Yes — the residual is not inert. The structure of $\mathbb{N} \subset \mathbb{Z}_2$ (terminating binary tails; finite description) adds three constraints beyond the four imported interfaces.

(i) Finite-seed determinism. Letters $b_0, \dots, b_n$ are determined by $x_0 \bmod 2^{B_n+1}$; once $B_n + 1 > \log_2 x_0$ the class pins x_0 uniquely and *every subsequent letter is forced* by the $\times 3$ carry cascade, with zero remaining freedom. The free phase lasts only $n^* \approx \frac{2}{3} \log_2 x_0$ steps (for $\{1,2\}$-words, $B_n \approx \frac{3}{2} n$). Hence the whole infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$, while repetition rigidity

forces near-maximal *factor* complexity in banded windows: the counterexample profile sharpens from "high-complexity calibrated pseudorandom word" to *self-generated pseudorandomness of a fixed finite carry transducer run from a finite seed* — logarithmic description length, maximal local diversity. In scheduler-adversary terms: the adversary's entire budget is $\log_2 x_0$ bits, spent in the first two-thirds-of-$\log_2 x_0$ steps; afterwards it makes no choices. Measured on the survival champions:

n_0	27	4591	31911	517191	687871
survival D	17	27	33	38	50
forced survival	13	16	22	24	33
forced fraction	76%	59%	67%	63%	66%

Two-thirds of a champion's life is carry-generated, not chosen.

(ii) Countability. The language's calibrated, shadow-biased, high-complexity word class is a Cantor continuum; integer-realizable words form a *countable, computable* subfamily — one word per odd integer, all emitted by the same finite-state machine ($\times 3+1$ in base 2, LSB-first, is a bounded-carry transducer). Concretely: of the 2^{30} words of depth 30 over $\{1, 2\}$, just 66 are realized by an integer below 10^6. Machine-style attacks (the LTE round chain, the 3-smooth cascade) are therefore not special cases but the *general* case: every integer is a finite seed of the same automaton.

(iii) A fifth interface. In the forced regime the letters read binary digit streams of 3-power multiples of the seed, so effective digit-counting theorems (Stewart: the number of nonzero binary digits of 3^n grows effectively) become applicable imports against late-time ghost alignment — a candidate fifth archimedean interface beyond clocks, counting, drift, and Baker.

Caveat. Constraints (i)–(ii) carry no measure-theoretic exclusion power — one cannot Borel–Cantelli a single computable point — which is consistent with the wall being pointwise. Their value is structural: they shrink the *profile* (what a counterexample must be) from a continuum of conspiracies to the output set of one finite machine, and they certify that automaton-level methods address the general case.

Verdict

Faithful compression = lossless coding + provable completeness on the declared categories + explicitly isolated residual. All three hold: the conjugacy (Theorem 1), the finite-modulus completeness theorems (Theorems 2 and 3, with null hidden-channel tests), and the residual — $\mathbb{N}$-positivity, the ghost gap — which is not compressed away but stated as the open core, where every known framework independently arrives. What would falsify faithfulness is specific and testable: a statistic outside the language that predicts survival beyond its CRT projection. We looked; there is none so far, and the test suite ships with the program.

References

[1] J. C. Lagarias, *The $3x + 1$ problem and its generalizations*, and D. Bernstein, *A non-iterative 2-adic statement of the $3N + 1$ conjecture* (conjugacy).

[2] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi (2022).

[3] E. Y. Chang, arXiv:2603.11066v6 (2026), audited in the ghost-clock companion note.

[4] J. A. Janik, *Finite Spectral Shadows for the Collatz Valuation Cocycle* (2026), and companion notes (two clocks; universal ghost clock; narrow-band; CIC round chain).